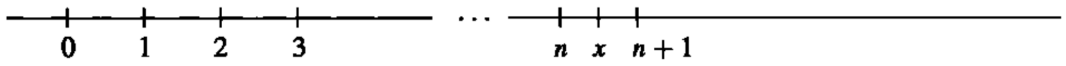
$\mathbb{N}$ is an unbounded subset of $\mathbb{R}$

Last time we gave $\mathbb{N}$ as an example of an unbounded set. We are now going to *prove* that $\mathbb{N}$ is unbounded. One might argue that this is "obvious" from the geometrical picture, as the real numbers look like



with every real number x between two integers n and $n+1$ (unless x itself is an integer). However, this picture contains an assumption: that if you place unit segments end-to-end you will eventually get a segment larger than any given segment. This assumption requires proof.

1. We prove that $\mathbb{N}$ is unbounded by contradiction. Suppose $\mathbb{N}$ were bounded. Then, since $\mathbb{N}$ is a nonempty subset of real numbers, it has a least upper bound. Then

$$n \leq \sup \mathbb{N} \text{ for all } n \in \mathbb{N}.$$

   (a) Argue that there exists some $m \in \mathbb{N}$ such that

$$m > \sup \mathbb{N} - 1. \tag{1}$$

   (b) Show that (1) leads to a contradiction.

   (c) What do you conclude?

2. Prove that for every $\varepsilon > 0$ there is a natural number n such that $\frac{1}{n} < \varepsilon$.
(*Hint:* Use Question 1.)

3. Let $A = \{\frac{1}{n} : n \in \mathbb{N}\} = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots\}$. Prove that $\inf A = 0$.

Rationals between reals

The next theorem states that there is a rational number between every pair of real numbers. Another way of saying this, is that if one picks any real number x, then within an arbitrary distance $\varepsilon > 0$, one can find a rational number r.

Theorem. For every two real numbers a and b with $a < b$, there exists a rational number r satisfying $a < r < b$.

Proof. First consider the case where $0 \leq a < b$. To prove the theorem, we need to find $m, n \in \mathbb{N}$ such that $a < \frac{m}{n} < b$, or, equivalently, [1]

$$na < m < nb. \tag{2}$$

4. Argue that there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < b - a. \tag{3}$$

5. Show that the inequality in (3) is equivalent to

$$na + 1 < nb. \tag{4}$$

6. Now argue that there exists a natural number greater than the real number na.

[1] For those worried about the directions of the inequalities in going from $a < \frac{m}{n} < b$ to $na < m < bn$: Since $0 < a < \frac{m}{n}$, m and n must have the same sign. Note that we are free to choose m and n to both be positive (otherwise, we just cancel the minus sign on top and bottom and call the resulting integers m and n). Since $n > 0$, we can multiply all sides of the inequality $a < \frac{m}{n} < b$ by the positive number n without reversing the inequalities to obtain (2).

7. Choose m to be the smallest positive integer greater than na. Choose m to be the least positive integer greater than na. [2] Show that

$$m - 1 \leq na < m, \tag{5}$$

thus proving the first inequality in (2). Explain why the first inequality is "$\leq$" rather than "$<$".

8. The first inequality in (5) implies

$$m \leq na + 1. \tag{6}$$

Combine this with the inequality in (4) to show that $m < nb$, completing the proof for the special case $0 \leq a < b$.

9. Now suppose $a < b \leq 0$. Use the special case just proved to show that there is a rational number between a and b.

10. The final case to check is $a < b$ with $a < 0$ and $b > 0$. However, in this case there is an obvious rational number between a and b. What is it?

[2] Note that m exists by the *Well-ordering property* of $\mathbb{N}$, which says that every nonempty subset of $\mathbb{N}$ has a least element.

Irrationals between reals

Theorem. Given any two real numbers a and b with $a < b$, there exists an irrational number x satisfying $a < x < b$. (*Hint*: Consider the real numbers $a - \sqrt{2}$ and $b - \sqrt{2}$.)